



Assignment Overview: Design the Data Architecture

Estimated time: 4 minutes

Learning objectives

- Identify key criteria for designing the data architecture for the merged organization, FutureMart.
- List the main tasks and expected deliverables for completing the assignment on designing the data architecture.

Introduction

In phase 1, you've thoroughly evaluated the existing data architectures of FashionMart and TrendyThreads. During your evaluation, you've determined potential gaps, inefficiencies, and improvement areas in the data architecture before designing the data architecture for the newly merged company, FutureMart. You've analyzed and documented the current data architecture of both companies and captured information about data storage, data pipelines, and integration points. You've reviewed data integration mechanisms, identified areas that need standardization, assessed the quality of existing data quality processes, observed security processes, compliance levels, data governance frameworks, and policies, and provided best practices for simplifying the architecture for future integration.

In this phase, you will work to design the data architecture of the merged organization, FutureMart.

Phase 2: Design the data architecture of the FutureMart

In Phase 2, you'll design a typical data architecture for a merger company, FutureMart, considering the evaluation report generated in Phase 1. You'll create the data architecture diagrams for FutureMart's new databases, entity-relationships (ER) diagrams, and table designs. You'll also develop initial schemas

and sketch out how the latest architecture fills gaps, bottlenecks, and previously noted inefficiencies.

References

Download the template below to design the data architecture of FutureMart.

[Template_Design the data architecture of a merged entity.docx](#)

Note: Right click on the link to open it in a new tab.

Tasks

Task 1: Create the enterprise data architecture (EDA) blueprint for FutureMart

In this task, you'll design a comprehensive EDA blueprint for FutureMart, including key capabilities to enhance a competitive workplace in the digital market.

In this task, you'll create a detailed EDA blueprint that includes the following layers, tools, and processes.

- Data sources
- Data integration
- Data storage
- Data processing
- Inventory and supply chain management
- Analytics and reporting
- Personalization and engagement
- AI-based shopping recommendations
- Real-time reporting
- Data governance

Task 2: Evaluate FutureMart's data architecture

Evaluate FutureMart's data architecture by analyzing its key components and identifying its strengths and areas for improvement.

1. Analyze key architectural components

In this task, you will assess the following key components:

- Data silos
- Legacy systems
- Scalable data architecture
- Data integration capabilities
- Robust data governance
- Performance bottlenecks in data processing
- High availability and disaster recovery plans
- Analytics and BI tools data security measures

2. Document strengths and areas for improvement

In this task, you'll identify strengths and areas for improvement in FutureMart's data architecture.

Task 3: Designing the FutureMart's OLTP database and ER diagram

In this task, you'll create an entity-relationship diagram (ERD) and define the database structure for FutureMart's OLTP system.

1. Define database requirements

In this task, you'll record sales for all apparel and track inventory levels for FutureMart's OLTP system.

2. Design the ER diagram

For creating an ER diagram for FutureMart:

- Identify key entities, such as customers, orders, products, and inventory.
- Define relationships between customers and orders and orders and products.
- Specify primary and foreign keys, and attributes for each entity.
- Ensure data normalization to reduce redundancy and improve efficiency.

Deliverables

In this reading, you have completed the following deliverables:

- A blueprint diagram of the **Enterprise Data Architecture (EDA)** for **FutureMart**.
- An assessment report highlighting the **strengths** and **areas for improvement** in **FutureMart's** data architecture.